

Technology Stack

Date	15 july 2026
Team ID	
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose

architecture and provide a brief justification for your choice.

Technology Stack Details

1	Frontend / Client-Side	React.js, HTML, CSS, Bootstrap	React provides reusable components and dynamic UI rendering; Bootstrap ensures responsive design for all devices.
2	Backend / Server-Side	Python (Flask / FastAPI)	Python offers strong support for AI/NLP libraries and APIs; Flask/FastAPI ensures fast processing and easy integration with ML models.
3	Database / Data Storage	MongoDB	Handles semi-structured data like transcripts and feedback efficiently; flexible schema supports evolving project needs.

4	Cloud / Hosting / Deployment	Microsoft Azure (App Service + Blob Storage)	Provides scalable hosting, secure data storage, and easy integration with AI services.
5	Version Control & CI/CD	GitHub + GitHub Actions	Enables seamless collaboration, version tracking, and automated testing/deployment pipelines.
6	Third-Party APIs / Other Tools	OpenAI Whisper API, HuggingFace Transformers, Azure Cognitive Services	Whisper handles speech-to-text conversion; Transformers enable semantic similarity analysis; Azure ensures secure and reliable AI integration.